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Department of Computer Science

CS11021– Programming Laboratory (Intake 43)

Lab Sheet 8

1. Develop a C++ program that takes a fixed monthly saving amount and calculates the total savings after **n months**.
2. A pedestrian signal alternates between **Walk** and **Stop** every *y* seconds.
Develop a C++ program to display the signal status for each second over a **30-second period**.
3. A vehicle's fuel consumption decreases by **3% every year** due to engine wear.
Develop a program to calculate and display the fuel efficiency for the next **8 years**.
4. Develop a C++ program that accepts **daily rainfall values for 5 days** and calculates the **total rainfall and average rainfall**.
5. A library has **n shelves**.
Develop a program to print shelf numbers but **skip shelves that are multiples of 7**.
6. Develop a C++ program to display a **classroom seating arrangement**, where rows represent benches and columns represent seat numbers.
7. Develop a C++ program that simulates a **countdown timer** starting from a given number of seconds and prints the remaining time.
8. A shop has **n customers**, and each customer buys **m items**.
Develop a program that takes item prices and calculates the **total bill and average item price** for each customer.
9. Generate the following patterns using stars (*). Refer to Figure 01.
10. Develop a C++ program to find and print all even numbers between 1 and 200, and also count how many even numbers are found.
11. Develop a program that takes a word as input and counts the number of uppercase letters, lowercase letters, and digits.
12. Develop a program that prints numbers from **1 to 100** and:
 - Prints **“Even”** if the number is even
 - Prints **“Odd”** if the number is odd

- Prints “**Special**” if the number is divisible by both 4 and 6

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